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Subject - Artificial Intelligence
Code - OE-EC 804A

1) Describe about three cognitive skill on which AI focuses.

Ans - (i) learning - Refers to the ability of Machines to acquire Knowledge from data and experience
→ Machine Analyze data patterns
→ Algorithms improve performance overtime
→ uses Machine learning models
ex → email spam detection
→ Speech Recognition.

(ii) Reasoning - It is the ability of AI System to draw conclusions from available information using logical Rules
→ Uses knowledge base + inference engine
→ Helps in decision Making
→ Used in expert Systems.
ex - Fraud detection
→ chess playing AI

(iii) Self Correction - means AI System learn from mistakes and improve performance automatically.

2) ~~Describe~~ Differentiate between the augmented intelligence and artificial intelligence with example.

Artificial Intelligence	Augmented Intelligence
→ Machine simulate human intelligence and perform tasks automatically	→ AI assists human in decision Making
→ Minimal human involvement.	→ Human and AI work together
→ Replace human tasks	→ Enhance human capabilities
→ Machine Makes decisions	→ human makes final decision
→ ex - Autonomous Robots in Manufacturing.	→ AI System Medical Scan but doctor gives final diagnosis.

3) Discuss in details about structure of agent in AI?

Ans: AI agent is an entity that perceives the environment and take ~~ag~~ actions to achieve goals

An agent is anything that perceives through sensors and acts through actuators.

Basic Structure:

Agent = Architecture + Program

Components

i) environment
ii) Sensors.
iii) Agent program
iv) Actuator

example - Camera, Microphone, ~~Agent program~~ wheels, Speakers

Agent working process

environment → Sensors - Agent program → Actuator → env

4) Differentiate between Agglomerative and divisive clustering

Agglomerative clustering	Divisive clustering
→ Bottom up	→ Top down
→ less complex	→ More complex
→ Each data point is it own cluster	→ All data points in one cluster
→ clusters are merged step by step	→ clusters are split step by step
→ commonly used.	→ less commonly used

5) Discuss about any two metrices which are used to evaluate the ~~clusters~~ clusters in AI. specify any two application of clustering.

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Ans- 1) Silhouette Score

Silhouette Score measures how similar an object is to its own clusters compared to other clusters.

Range : -1 to 1

Interpretation

+1 → well clustered
0 → overlapping clusters
-1 → wrong clustering

Formula.

$$Score = \frac{(b-a)}{MAX(a,b)}$$

a → distance within cluster
b → distance to nearest cluster.

2) Davies Bouldin index

it evaluates cluster based on:
→ Cluster compactness
→ Distance between clusters

interpretation :

- lower value → better clustering

Application of Clustering

→ customer segomentation
→ image Segmentation
→ Document Classification
→ Anomaly Detection.